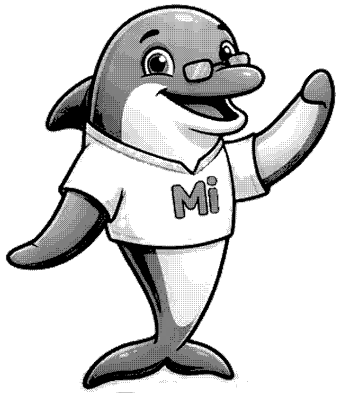
Math Inquirers

Extension Workbook

with answers

Level 4

Self-study | Ages 9-11



My name _____

About this book

Math Inquirers Extension Workbook, Level 4
Self-study practice with answers.

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Good to know

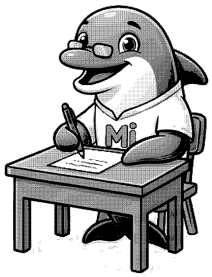
- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

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Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- The rulers in Unit 9 are printed at real size.

The four page names

- **Start here.** The main path. Do these first.
- **Think harder.** Longer problems. Hints are printed upside down.
- **Go further.** A weekend page. Big challenges, just for fun.
- **Check yourself.** Four quick questions at the end.

Contents

	About this book Copyright and licence	page 2
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1	**Factor hunters** Not in this proof edition	–
2	**Big number puzzles** Not in this proof edition	–
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6	**Angle Turns** Angle turns · follows Holiday Program Lesson 6 · answers on page 16	page 6
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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	**Factor hunters**	Not in this proof edition				☐
2	**Big number puzzles**	Not in this proof edition				☐
3	**Operation detectives**	Not in this proof edition				☐
4	**Fraction frontier**	Not in this proof edition				☐
5	**Decimal detectives**	Not in this proof edition				☐
6	**Angle Turns**	6-7	8-9	10 and 12	11	☐
7	**Area architects II**	Not in this proof edition				☐
8	**Symmetry and nets**	Not in this proof edition				☐
9	**Data detectives**	Not in this proof edition				☐
10	**Puzzle day**	Not in this proof edition				☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

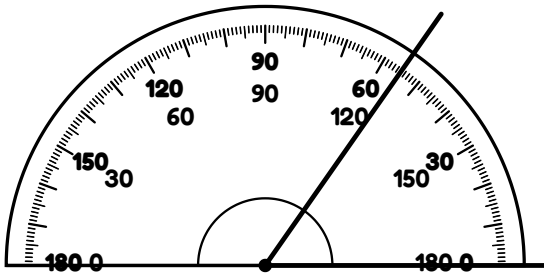
Learn it: Angles

My goal. I can measure angles and find missing angles. I can find the angle between clock hands and turn on the 8-point compass.

You need a 180° protractor and a ruler for this unit. A full turn is 360°, a half turn is 180° and a quarter turn is 90°. To measure, count up from the 0 on one arm.

Worked example 1

How big is the angle? Use the scale that starts at 0 on the arm.

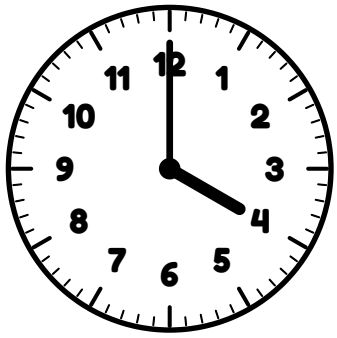


1. **Put the centre of the protractor on the corner. Put the 0 line on one arm.**
2. **Read the scale that starts at 0 on that arm.**

Answer: The angle is 55°.

Worked example 2

What is the angle between the clock hands at 4 o'clock?



1. **A clock has 12 equal spaces in 360°. 360 ÷ 12 = 30, so 1 space is 30°.**
2. **At 4 o'clock the hands are 4 spaces apart: 4 × 30°.**

Answer: The angle is 120°.

How to measure

1. Estimate first. Is it less than 90° or more?
2. Centre on the corner. 0 line on one arm.
3. Read from the 0 on that arm. Check with your estimate.

Good to know

- Compass: 1 step = 45°.
- Clock: 1 space = 30°.
- Minute hand: 1 minute = 6°.

Your turn

Answer at the back of the book.

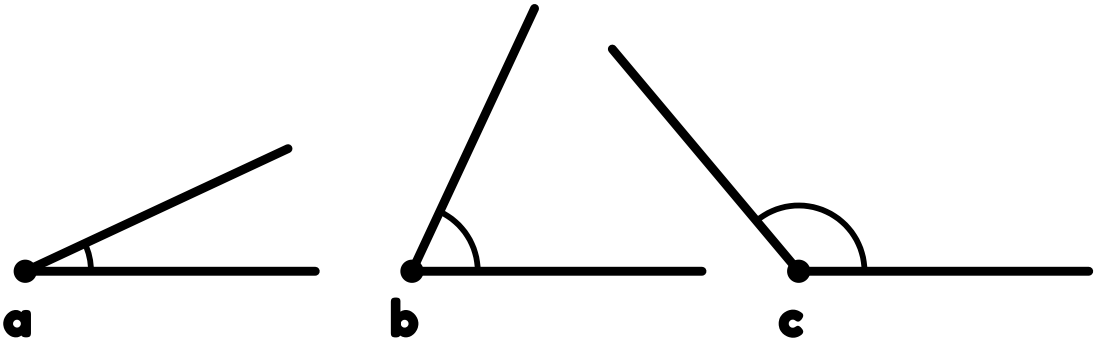
Try **Ali faces NE. He turns 135° clockwise. Which way does he face now?**

E SE S SW

Start here

Do these first. Take your time.

1 **Estimate each angle. Then measure it with your protractor. The arms are short, so draw them longer with a ruler first.**



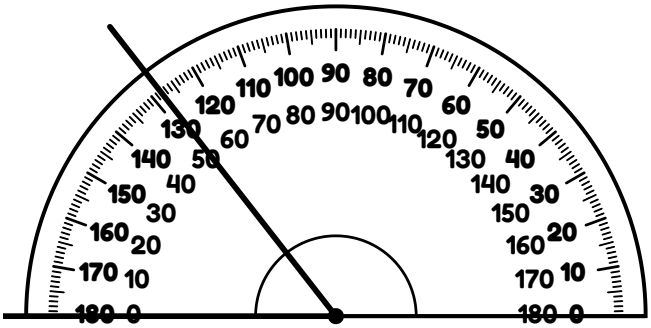
(a) a = ☐ ° (b) b = ☐ ° (c) c = ☐ °

2 **Read the angle from the protractor.**

(a) The angle is ☐ °

(b) Is it smaller or greater than a right angle?

smaller greater



3 **Four rays OA, OB, OC and OD make 3 angles side by side. Find each angle.**

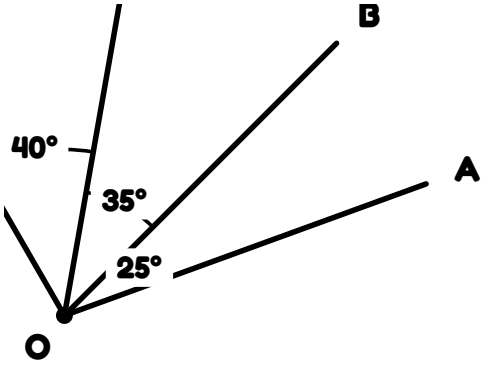
(a) ∠AOC = ☐ °

(b) ∠BOD = ☐ °

(c) ∠AOD = ☐ °

(d) How many angles are there in all?

☐



Start here

Keep going. Check your work.

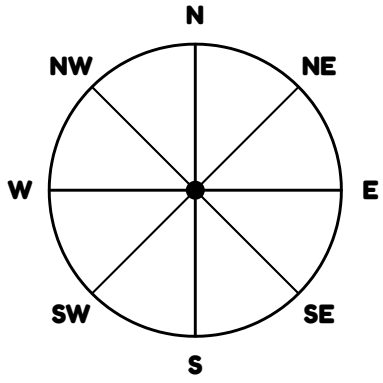
4 **One step from a direction to the next is 45°. Find the direction.**

(a) Ali faces W. He turns 90° clockwise. He now faces ☐

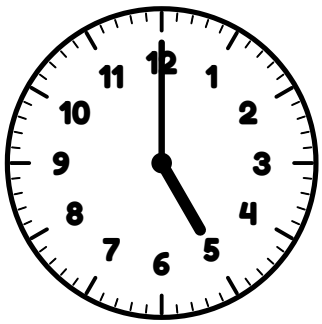
(b) Mei faces SE. She turns 180°. She now faces ☐

(c) Siti faces NE. She turns 270° clockwise. She now faces ☐

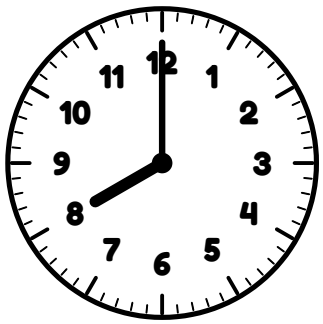
(d) Ravi turns 135° anticlockwise and now faces S. At first he faced ☐



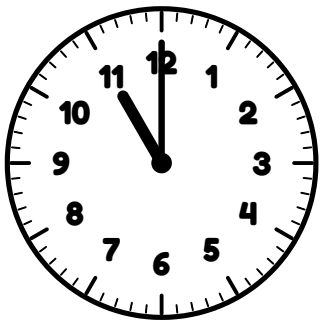
5 **Find the smaller angle between the clock hands.**



5 o'clock



8 o'clock



11 o'clock

(a) 5 o'clock: ☐ ° **(b)** 8 o'clock: ☐ ° **(c)** 11 o'clock: ☐ °

6 **Ben faces W. He wants to face NE. What is his smallest turn?**

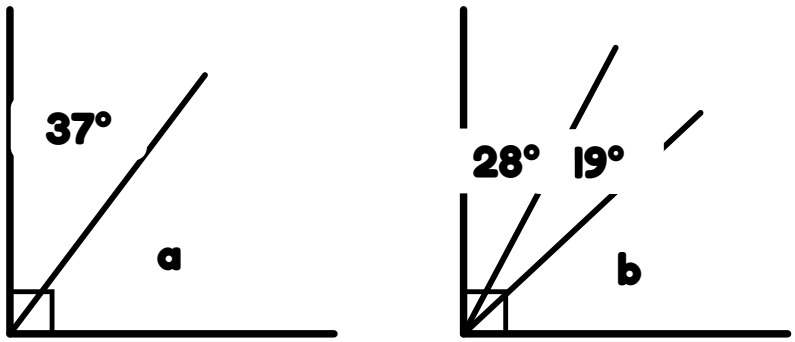
(a) The turn is ☐ °

(b) The direction is clockwise anticlockwise

Start here

Nearly there. You can do it.

7 **A corner of a rectangle is 90°. A line splits it. Find the angles a and b. The drawings are not to scale.**



(a) a = ☐ ° **(b)** b = ☐ °

8 **A robot faces E. It turns 135° anticlockwise. Then it turns 90° clockwise. Then it turns 45° clockwise.**

(a) It now faces ☐

(b) In all it turned ☐ °

9 **A corner of a rectangle is split into 2 angles. One angle is 5 times the other.**

(a) The smaller angle is ☐ °

(b) The greater angle is ☐ °

Draw a bar model here.

10 **How long does the minute hand take to turn through each angle?**

(a) 90°: ☐ minutes **(b)** 180°: ☐ minutes

(c) 270°: ☐ minutes **(d)** 60°: ☐ minutes

Think harder

Hints are upside down. Try first!

1 **Beyond Level 4: Angles on a line and at a point**
Angles on a straight line add up to 180°. Angles around a point add up to 360°.

(a) On a straight line the angles are 47°, 85° and a. a = ☐ °

(b) Around a point the angles are 95°, 120°, 78° and b. b = ☐ °

Working: _____

_____

Hint. Add the angles you know. Then take the sum away from the total.

2 **Find the smaller angle between the clock hands.**

(a) At 7:30 it is ☐ °

(b) At 1:30 it is ☐ °

Working: _____

_____

_____

Hint. At half past, the hour hand is halfway between 2 numbers.

3 **A robot faces N. It turns 90° clockwise, then 135° anticlockwise, then 225° clockwise, then 45° anticlockwise.**

(a) It now faces ☐

(b) In all it turned ☐ °

Working: _____

_____

Hint. Count the steps of 45°. Clockwise is plus, anticlockwise is minus.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Six rays come out of one point and make 5 equal angles side by side. The angle between the 2 outside rays is 150°.**

(a) How many angles can you find in all? ☐

(b) How many of them are right angles? ☐

Working: _____

Hint. One small angle is 30°. Count the angles made of 1 part, 2 parts, and so on.

2 **A robot faces N. It turns 135° clockwise again and again.**

(a) After how many turns does it first face N again? ☐

(b) After 50 turns it faces ☐

List: _____

Hint. 135° is 3 steps of 45°. Write the directions after each turn.

3 **A corner of a rectangle is 90°. It is split into 3 angles. The middle angle is twice the smallest. The greatest is 6 times the smallest.**

(a) The smallest angle is ☐ °

(b) The greatest angle is ☐ °

Working: _____

_____

Hint. Count the equal parts. 1 + 2 + 6 = 9.

4 **At which o'clock times (from 1 o'clock to 12 o'clock) is the smaller angle between the clock hands more than 100° and less than 160°?**

(a) How many times are there? ☐

(b) Write the times: ☐ ☐ ☐ ☐

Hint. Write the angle for every o'clock time first.

✔ Check yourself

No hints. Try each one on your own.

1 **Find the smaller angle between the clock hands at 7 o'clock.** ☐°

Working: _____

2 **Aisha faces S and turns 135° clockwise. Which way does she face now?** ☐

Working: _____

3 **A corner of a square is split into 2 angles. One angle is 41°. How big is the other?**

☐°

Working: _____

4 **How many degrees does the minute hand turn from 5:10 to 5:35?** ☐°

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Factor hunters	☐	☐	☐	☐	/ 4	
2	Big number puzzles	☐	☐	☐	☐	/ 4	
3	Operation detectives	☐	☐	☐	☐	/ 4	
4	Fraction frontier	☐	☐	☐	☐	/ 4	
5	Decimal detectives	☐	☐	☐	☐	/ 4	
6	Angle Turns	☐	☐	☐	☐	/ 4	
7	Area architects II	☐	☐	☐	☐	/ 4	
8	Symmetry and nets	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 4.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 6 · Angle Turns

Your turn
page 6

Answer: S
Why: 135° is 3 steps of 45°. Clockwise from NE: E, SE, S. He faces S.

START HERE

1
page 7

Answer: (a) 25 (b) 65 (c) 130
Why: Each angle starts at 0 on the horizontal arm. a = 25°, b = 65° and c = 130°. Answers within 2° are fine. Angle c is bigger than a right angle, so a good estimate is more than 90°.

2
page 7

Answer: (a) 52 (b) smaller
Why: One arm lies along the baseline at 0 on the left. Read the scale that starts at 0 on the left arm: 52°. A right angle is 90°, and 52° is smaller. The other scale says 128°, which is the angle measured from the other arm.

3
page 7

Answer: (a) 60 (b) 75 (c) 100 (d) 6
Why: The angles add. ∠AOC = 25° + 35° = 60°. ∠BOD = 35° + 40° = 75°. ∠AOD = 25° + 35° + 40° = 100°. The angles are AOB, BOC, COD, AOC, BOD and AOD: 6 angles.

4
page 8

Answer: (a) N (b) NW (c) NW (d) NW
Why: (a) 90° is 2 steps. From W clockwise: NW, N. (b) 180° is 4 steps. From SE: S, SW, W, NW. (c) 270° is 6 steps. From NE clockwise: E, SE, S, SW, W, NW. (d) Go back the other way: 135° is 3 steps clockwise from S: SW, W, NW.

5
page 8

Answer: (a) 150 (b) 120 (c) 30
Why: Each space between numbers is 30°. At 5 o'clock the hands are 5 spaces apart: 150°. At 8 o'clock they are 4 spaces apart the short way: 120°. At 11 o'clock they are 1 space apart: 30°.

6
page 8

Answer: (a) 135 (b) clockwise
Why: Clockwise from W: NW, N, NE is 3 steps, 135°. Anticlockwise from W: SW, S, SE, E, NE is 5 steps, 225°. The smallest turn is 135° clockwise.

7
page 9

Answer: (a) 53 (b) 43
Why: (a) a + 37° = 90°, so a = 90° − 37° = 53°. (b) The three angles add to 90°: b = 90° − 19° − 28° = 43°.

8
page 9

Answer: (a) E (b) 270
Why: 135° anticlockwise from E: NE, N, NW. 90° clockwise from NW: N, NE. 45° clockwise from NE: E. It faces E. The turns add to 135° + 90° + 45° = 270°.

9
page 9

Answer: (a) 15 (b) 75
Why: The corner is 90° and is made of 1 + 5 = 6 equal parts. 90° ÷ 6 = 15° for 1 part. The greater angle is 5 × 15° = 75°.

10
page 9

Answer: (a) 15 (b) 30 (c) 45 (d) 10
Why: The minute hand turns 6° each minute. 90° ÷ 6 = 15. 180° ÷ 6 = 30. 270° ÷ 6 = 45. 60° ÷ 6 = 10.

THINK HARDER

1 page 10
Answer: (a) 48 (b) 67
Why: (a) 47° + 85° = 132°, and 180° − 132° = 48°. (b) 95° + 120° + 78° = 293°, and 360° − 293° = 67°.

2 page 10
Answer: (a) 45 (b) 135
Why: At 7:30 the minute hand is on the 6. The hour hand is halfway between 7 and 8, which is 1 and a half spaces from the 6: 45°. At 1:30 the hour hand is halfway between 1 and 2, which is 4 and a half spaces from the 6: 4 × 30° + 15° = 135°.

3 page 10
Answer: (a) SE (b) 495
Why: In steps of 45°: +2, −3, +5, −1. From N: 2 steps is E, then 3 steps back is NW, then 5 steps forward is S, then 1 step back is SE. The turns add to 90° + 135° + 225° + 45° = 495°.

GO FURTHER

1 page 11
Answer: (a) 15 (b) 3
Why: Each small angle is 150° ÷ 5 = 30°. Angles made of 1 part: 5. Of 2 parts: 4. Of 3 parts: 3. Of 4 parts: 2. Of 5 parts: 1. In all 5 + 4 + 3 + 2 + 1 = 15. A right angle is 3 parts of 30°, and there are 3 of them.

2 page 11
Answer: (a) 8 (b) W
Why: Each turn is 3 steps. The directions are SE, W, NE, S, NW, E, SW, N. It faces N again after 8 turns. 50 turns is 6 lots of 8, which is 48 turns, and 2 more turns. After 48 turns it faces N. 2 more turns take it to SE and then W.

3 page 11
Answer: (a) 10 (b) 60
Why: The 3 angles are 1, 2 and 6 equal parts. That is 9 parts in all, and 9 parts = 90°, so 1 part = 10°. The smallest is 10°, the middle is 20° and the greatest is 60°.

4 page 11
Answer: (a) 4 (b) 4, 5, 7 and 8 o'clock
Why: The angle at 1 to 12 o'clock is 30°, 60°, 90°, 120°, 150°, 180°, 150°, 120°, 90°, 60°, 30°, 0°. Those between 100° and 160° are 120°, 150°, 150° and 120°, at 4, 5, 7 and 8 o'clock.

CHECK YOURSELF

1 page 12
Answer: 150
Why: At 7 o'clock the hands are 5 spaces apart the short way. 5 × 30° = 150°.

2 page 12
Answer: NW
Why: 135° is 3 steps of 45°. Clockwise from S: SW, W, NW.

3 page 12
Answer: 49
Why: The corner is 90°. 90° − 41° = 49°.

4 page 12
Answer: 150
Why: From 5:10 to 5:35 is 25 minutes. 25 × 6° = 150°.